Global parameters

- Target `spark`, `use_gpu` on
- Quality: `max_depth` 5, `samples` 10^6, `max_paths_per_tx` 10^5, `seed` 42
- `los` + `specular_reflection` + `refraction` + `diffraction` on, `diffuse_reflection` off
- `cfr_points` 0, no measure surface
- Band `lora_915` (915 MHz)
- Antenna preset `iso_reference` (TX iso V, RX iso V)
- Sensors: role TX (`power_dbm` 14), 1 sensor per plant 1 cm from stem, leaf or tray, `sensor_seed` 1
- Access point: role RX, 2 m above the trays, no antenna attributes on the points
- No moisture attribute: leaf uses the library default (80 %)

Batch 1: growth stage × species (20 work items)

x = growth stage, y = species.

stage 0, strawberry	stage 1, strawberry	stage 2, strawberry	stage 3, strawberry	stage 4, strawberry
stage 0, basil	stage 1, basil	stage 2, basil	stage 3, basil	stage 4, basil
stage 0, salad	stage 1, salad	stage 2, salad	stage 3, salad	stage 4, salad
stage 0, tomato	stage 1, tomato	stage 2, tomato	stage 3, tomato	stage 4, tomato

Batch 2: growth stage × frequency (20 work items, tomato only)

x = growth stage, y = frequency. Null: `frequency` = `@frequency`.

stage 0, 868 MHz	stage 1, 868 MHz	stage 2, 868 MHz	stage 3, 868 MHz	stage 4, 868 MHz
stage 0, 915 MHz	stage 1, 915 MHz	stage 2, 915 MHz	stage 3, 915 MHz	stage 4, 915 MHz
stage 0, 2.437 GHz	stage 1, 2.437 GHz	stage 2, 2.437 GHz	stage 3, 2.437 GHz	stage 4, 2.437 GHz
stage 0, 5.18 GHz	stage 1, 5.18 GHz	stage 2, 5.18 GHz	stage 3, 5.18 GHz	stage 4, 5.18 GHz

Batch 3: access point density × growth stage (20 work items, tomato only)

x = access point density, y = growth stage. AP wrangle: 1 AP per `@area_per_ap` m^2 (instead of the fixed 100); room $5.8 \times 5.8\,\mathrm{m} = 33.6\,\mathrm{m}^2$.

stage 0, 1 AP per 34 m^2	stage 0, 1 AP per 17 m^2	stage 0, 1 AP per 8 m^2	stage 0, 1 AP per 5 m^2
stage 1, 1 AP per 34 m^2	stage 1, 1 AP per 17 m^2	stage 1, 1 AP per 8 m^2	stage 1, 1 AP per 5 m^2
stage 2, 1 AP per 34 m^2	stage 2, 1 AP per 17 m^2	stage 2, 1 AP per 8 m^2	stage 2, 1 AP per 5 m^2
stage 3, 1 AP per 34 m^2	stage 3, 1 AP per 17 m^2	stage 3, 1 AP per 8 m^2	stage 3, 1 AP per 5 m^2
stage 4, 1 AP per 34 m^2	stage 4, 1 AP per 17 m^2	stage 4, 1 AP per 8 m^2	stage 4, 1 AP per 5 m^2

(= 1 / 2 / 4 / 7 access points)

Batch 4: moisture × growth stage (20 work items, tomato only)

x = moisture, y = growth stage.

stage 0, moisture 20	stage 0, moisture 50	stage 0, moisture 70	stage 0, moisture 90
stage 1, moisture 20	stage 1, moisture 50	stage 1, moisture 70	stage 1, moisture 90
stage 2, moisture 20	stage 2, moisture 50	stage 2, moisture 70	stage 2, moisture 90
stage 3, moisture 20	stage 3, moisture 50	stage 3, moisture 70	stage 3, moisture 90
stage 4, moisture 20	stage 4, moisture 50	stage 4, moisture 70	stage 4, moisture 90

Batch 5: occupancy × room size (15 work items, tomato, stage 4)

x = occupancy (people + robots), y = room size. `disturbance_seed` 1 fixed: each step keeps the movers of the previous step and adds more. Access points 1 per 11 m^2 in every room size (small 3, medium 9, large 36 APs).

small room, 0	small room, 1	small room, 2	small room, 4	small room, 8
medium room, 0	medium room, 1	medium room, 2	medium room, 4	medium room, 8
large room, 0	large room, 1	large room, 2	large room, 4	large room, 8

(small 5.8 × 5.8 m, medium 10 × 10 m, large 20 × 20 m)